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Privacy-Preserving Federated Learning for Distributed Skin Disease Classification

by
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Prof Michael Bruenig
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Dear Professor Bruenig,

In accordance with the requirements of the Degree of Bachelor of Computer Science (Honours) in the School of Electrical Engineering and Computer Science, I submit the following thesis entitled

"Privacy-Preserving Federated Learning for Distributed Skin Disease Classification"

The thesis was performed under the supervision of Dr Ruihong Qiu. I declare that the work submitted in the thesis is my own, except as acknowledged in the text and footnotes, and that it has not previously been submitted for a degree at the University of Queensland or any other institution.

Yours sincerely,

Alex Gu

ALEX GU

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This thesis is composed of my original work, and contains no material previously published, written by another person, or generated by artificial intelligence (AI), except where clearly referenced and acknowledged either in the main text or the preliminary pages of this thesis. I have clearly stated the contribution of others to jointly authored works that I have included in my thesis.

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Abstract

Skin disease classification using deep learning has shown strong potential but training accurate models requires large, diverse datasets often distributed across healthcare institutions. Privacy regulations such as the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) prevent centralised data aggregation. Prior work has shown that even sharing model updates rather than raw data can leak sensitive patient information. Federated learning offers a framework for collaborative training across distributed nodes without centralising data, but standard federated averaging lacks convergence guarantees. This project evaluates federated learning approaches for multi-class skin disease classification in a simulated distributed healthcare environment. A working dataset of approximately 20,000 images across 30 diagnostic classes, drawn from publicly available skin disease image collections, will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, evaluating classification accuracy, communication efficiency, and model behaviour under data that is not independently and identically distributed (non-IID). The findings of this project will provide guidance for selecting and configuring federated learning for medical image classification tasks in privacy-constrained distributed settings.

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Chapter 1

Introduction

Skin diseases are among the most common health conditions worldwide [1] and deep learning has demonstrated strong performance in skin disease classification from images [2]. However, training models that generalise well requires diverse data across patient demographics and imaging setups. In practice this data sits in separate hospitals and research groups, each with a relatively small local dataset. Privacy regulations including the Health Insurance Portability and Accountability Act (HIPAA) and the General Data Protection Regulation (GDPR) restrict the centralisation of patient data [3], preventing the ability to build robust classifiers through conventional centralised training.

Federated learning addresses this by enabling multiple institutions to collaboratively train a shared model without exchanging raw data. Each participant trains on its local dataset and shares only model updates with a central server, which aggregates them into a global model. McMahan et al. introduced Federated Averaging (FedAvg), which reduces communication costs by performing multiple local training epochs before aggregation [4]. However, FedAvg lacks convergence guarantees when data across participants is not independently and identically distributed (non-IID), a condition that is common in healthcare settings where different institutions may see different patient demographics and disease distributions. FedProx extends FedAvg by introducing a proximal term that constrains local model updates, providing convergence guarantees under heterogeneous conditions [5]. Furthermore, federated learning alone does not fully resolve privacy concerns. Prior work has shown that shared model updates can leak sensitive information, and additional mechanisms may be needed [3].

This project evaluates federated learning for multi-class skin disease classification in a simulated distributed healthcare environment. A working dataset of approximately 20,000 images across 30 diagnostic classes, drawn from publicly available skin disease image collections, will be partitioned across multiple nodes. FedAvg and FedProx will be implemented and compared against a centralised training baseline, evaluating classification accuracy and communication efficiency under non-IID data distributions.

The remainder of this thesis is organised as follows. Chapter 2 reviews the relevant

literature on federated learning, data heterogeneity, and privacy. Chapter 3 describes the methodology, including the dataset, centralised baseline, and federated learning configurations. Chapters 4 and 5 present the experimental studies, and Chapter 6 concludes.

Chapter 2

Background and Literature review

2.1 Skin Disease Classification with Deep Learning

Skin diseases are among the most prevalent health conditions globally, affecting hundreds of millions of people and contributing significantly to the global burden of disease [1]. Accurate and timely diagnosis is critical for effective treatment, but access to dermatological expertise is limited in many regions. This has motivated research into automated classification systems that can assist clinicians by analysing images of skin conditions.

Deep learning, particularly convolutional neural networks (CNNs), has demonstrated strong performance in this domain. Esteva et al. [2] showed that a CNN trained on a large dataset of clinical images could classify skin cancer at a level comparable to board-certified dermatologists, establishing deep learning as a viable tool for skin disease diagnosis. Since then, the field has benefited from the availability of large public datasets such as HAM10000, a collection of 10,015 dermatoscopic images spanning seven diagnostic categories [6]. These datasets have enabled accelerated progress in automated skin disease classification.

The remaining bottleneck is data access. Generalisation across skin tones and imaging hardware requires broader datasets than any single institution holds. In practice, such data is distributed across hospitals and associated health and research institutions, and privacy regulations prevent its centralisation.

2.2 Federated Learning

Federated learning offers a solution for training machine learning models across distributed data sources without centralising the data itself. McMahan et al. [4] introduced Federated Averaging (FedAvg) which frames the learning problem as minimising a global objective function across K clients:

$$\min_{w} f(w) = \sum_{k=1}^{K} \frac{n_k}{n} F_k(w) \tag{2.1}$$

where w are the model parameters, $F_k(w)$ is the local objective function for client k, n_k is the number of data points held by client k, and n is the total number of data points across all clients. Each client's contribution to the global objective is weighted by the proportion of data it holds. Figure 2.1 illustrates this process.

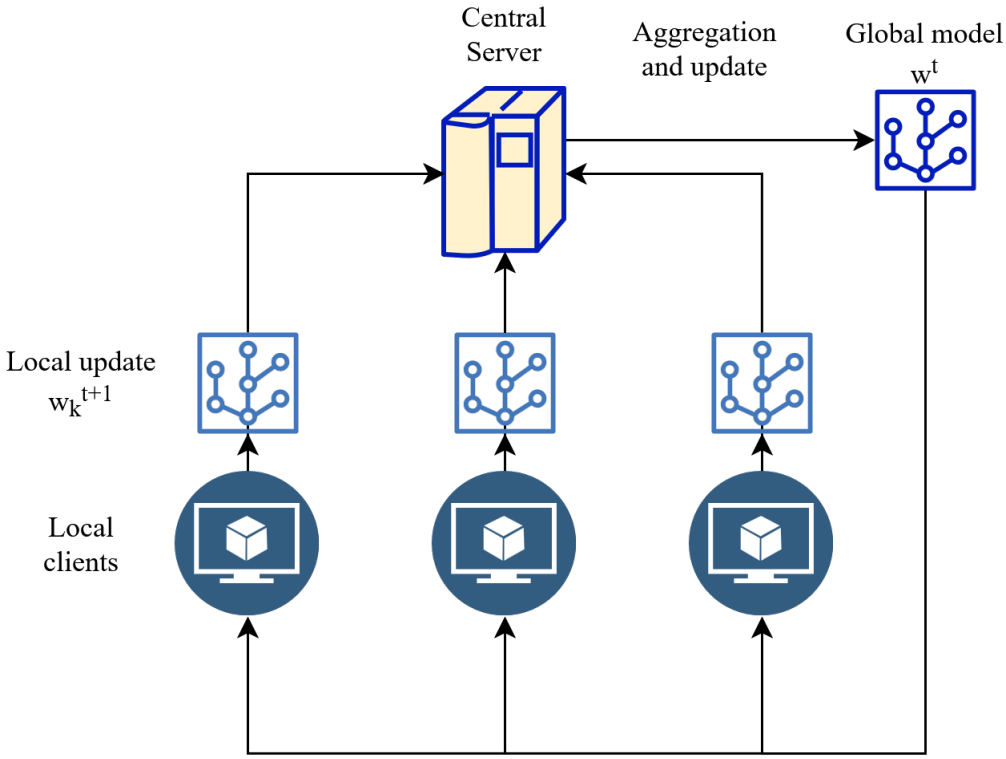


Figure 2.1: Overview of the federated learning pipeline

In each communication round, the server distributes the current global model w^t to a subset of clients. Each selected client initialises its local model with w^t and performs E epochs of stochastic gradient descent on its local data, producing updated parameters w_k^{t+1}. The server then aggregates these updates by computing a weighted average:

$$w^{t+1} = \sum_{k=1}^{K} \frac{n_k}{n} w_k^{t+1} \tag{2.2}$$

By performing multiple local training epochs before communication, FedAvg significantly reduces communication costs compared to approaches that require gradient updates after every training step.

Li et al. [7] surveyed the main challenges. 1) Communication costs are high when bandwidth is limited or models are large. 2) Participating devices vary in compute and connectivity. 3) Data across nodes is usually non-IID. 4) Shared updates can still leak information about training data.

Communication cost is the first of these challenges and has attracted its own line of work. Konečný et al. [8] proposed two ways to reduce communication cost: structured

updates, which learn each client update directly in a restricted low-rank or random-mask space, and sketched updates, which compute a full update and then compress it through quantisation, random rotation and subsampling before transmission. Sattler et al. [9] observed that earlier compression schemes either compressed only the updates or degraded under non-IID data and proposed sparse ternary compression, which combines top-k sparsification, ternarisation and Golomb encoding of the updates with a mechanism for compressing the downloading as well. A simpler reduction applies whenever a pre-trained backbone is partially frozen: parameters that do not change during local training carry no information and need not be transmitted. Each of these methods trades a small amount of accuracy or convergence speed for a reduction in bytes per round, and the appropriate trade-off depends on the bandwidth available to participating institutions.

2.3 Handling Non-IID Data in Federated Learning

Data distributions across participating nodes are rarely uniform. When each node holds data that is non-IID, local models trained on these subsets may diverge significantly from one another. Averaging such divergent models can produce a global model that performs poorly compared to one trained on centralised data.

The mechanism behind this degradation is client drift [10]. During the E local epochs of each round, a client's model moves toward the minimiser of its own local objective F_k rather than the global objective f. When the local label distribution differs from the global one, these two minimisers are far apart and the local update accumulates a bias in the direction of the local optimum. Drift grows with the number of local steps, the learning rate and the degree of label skew and shrinks when many clients are averaged per round or when the starting point is already close to a good solution for every client. The methods reviewed below differ in how they act on drift.

FedAvg provides no mechanism to limit drift. Li et al. [5] proposed FedProx, which penalises it directly by adding a proximal term to the local objective:

$$\min_{w} h_k(w; w^t) = F_k(w) + \frac{\mu}{2}\|w - w^t\|^2 \tag{2.3}$$

where $F_k(w)$ is the local loss, w^t is the current global model and μ controls the strength of the penalty. The term pulls each local model back toward the global model in proportion to how far it has moved. Its gradient $\mu(w - w^t)$ is zero at the start of local training and grows with drift. This modification provides convergence guarantees under heterogeneous data distributions, which FedAvg lacks. FedProx also accommodates systems heterogeneity by allowing nodes to perform variable amounts of local computation, since a client that completes only part of its local work is still anchored to w^t. When μ is small relative to the local learning rate the penalty is negligible and FedProx reduces to FedAvg; when it is large the penalty dominates the local loss and slows convergence.

Other approaches have been proposed to tackle non-IID issues. Karimireddy et al. [10]

introduced SCAFFOLD, which uses control variates to correct for the drift in local updates. This reduces the variance caused by heterogeneous data. Wang et al. [11] proposed Federated Matched Averaging (FedMA), which aligns neurons across local models before averaging. This prevents neurons in different local models from learning equivalent features in different positions. Wang et al. [12] proposed FedNova, which normalises local updates to account for differences in the number of local training steps, correcting an inconsistency in FedAvg that is amplified under heterogeneous conditions. While this project focuses on FedAvg and FedProx, these alternative approaches provide context for understanding the breadth of strategies available for handling non-IID data in federated settings.

To evaluate federated learning methods under heterogeneous conditions, a consistent experimental method is required for simulating non-IID data splits. A widely adopted approach, introduced by Hsu et al. [13], is to partition data across clients using a Dirichlet distribution over class labels, parameterised by a concentration parameter α. Lower values of α produce more skewed label distributions across clients while higher values approach IID. This provides a controllable and reproducible way to study model behaviour across a wide range of heterogeneous conditions.

The theoretical case for drift correction is stronger than evidence suggestes. Li et al. [14] benchmarked FedAvg, FedProx, SCAFFOLD and FedNova on nine datasets under label skew, feature skew and quantity skew partitions and found that no algorithm consistently outperformed the others, with the newer methods beating FedAvg by a clear margin in only a few settings. FedProx tracked FedAvg closely through training in many cases. Having tuned μ over $\{0.001, 0.01, 0.1, 1\}$, they found the best value was always small, so the proximal term had little effect and the two methods reached similar accuracy at a similar speed. They also found that the number of local epochs had a large effect on accuracy and that its optimal value was sensitive to the partition. Their results suggest that the advantage of a drift-correction method depends on whether the experimental setting produces enough drift for the correction to matter.

2.4 Privacy in Federated Learning

Federated learning avoids sharing raw data but this alone does not guarantee privacy. Kaissis et al. [3] noted that the model updates transmitted during training can be exploited to infer information about the underlying data. Model inversion attacks use model outputs or gradients to reconstruct approximations of training inputs [15]. Membership inference attacks determine whether a specific data point was in the training set by analysing the model's response to that input [16].

Several defence mechanisms have been proposed to strengthen privacy in federated learning. Bonawitz et al. [17] designed a practical secure aggregation protocol using pairwise masking to protect each participant's contribution. This enables the server to compute the sum

of client updates without observing any individual update. This protocol is robust to client dropout, making it suitable for real-world deployments. Differential privacy provides a complementary approach by adding calibrated noise to the data or model updates to limit the information that can be inferred about any individual training example. Dwork et al. [18] established the formal foundations of differential privacy, while Abadi et al. [19] introduced DP-SGD, a training procedure that clips per-example gradients and adds noise during optimisation which enables deep learning models to be trained with formal privacy guarantees. These mechanisms can be combined with federated learning to provide stronger privacy protections, though they often introduce trade-offs with model accuracy.

2.5 Federated Learning in Medical Imaging

Federated learning has attracted significant interest in medical imaging, where data is inherently distributed across institutions and subject to strict privacy requirements. Rieke et al. [20] outlined the potential of federated learning to enable collaborative model development across hospitals without centralising patient data, identifying it as a key enabler for the future of digital health. They noted that the heterogeneity of clinical data, arising from differences in imaging equipment, protocols, and patient demographics makes the non-IID problem particularly important in this domain.

Sheller et al. [21] demonstrated that federated learning could be applied to brain tumour segmentation across multiple institutions. They achieved performance comparable to centralised training while keeping patient data local. Their work provided early evidence that federated learning is viable in medical imaging tasks with complex, heterogeneous data. More recently, Dayan et al. [22] conducted one of the largest federated learning studies to date, training a model to predict clinical outcomes in COVID-19 patients across 20 institutions worldwide. Their model outperformed any single-institution model, which justifies federated learning in practice.

2.6 Summary and Knowledge Gap

The preceding sections have reviewed the relevant literature across skin disease classification, federated learning, non-IID data handling, privacy mechanisms, and federated learning in medical imaging. Several key gaps emerge from this review:

1. **Limited evaluation of federated learning for multi-class skin disease classification.** Federated learning has been applied successfully in medical imaging, but primarily in radiology and pathology tasks with binary or few-class classification. Its performance on fine-grained, multi-class skin disease tasks with 20 or more classes has not been systematically studied.

2. **Insufficient comparison of federated algorithms under realistic non-IID conditions.** While FedProx was designed to address the non-IID limitations of

FedAvg, direct comparisons between the two on medical imaging tasks with realistic heterogeneity remain scarce. Most evaluations use standard benchmarks rather than domain-specific datasets where class imbalance and label skew occur.

3. **Lack of practical guidance on federated learning trade-offs for clinical imaging.** Existing studies report accuracy but often do not systematically evaluate communication efficiency or per-class performance alongside non-IID sensitivity. Understanding these trade-offs is essential for informing whether federated learning is practical for real-world deployment.

This project addresses these gaps by evaluating FedAvg and FedProx on a 30-class skin disease dataset to simulate realistic data heterogeneity. By comparing these approaches against a centralised baseline on classification accuracy, communication efficiency, and behaviour under non-IID distributions, this project aims to provide practical insight into the suitability of federated learning for fine-grained medical image classification.

Chapter 3

Methodology

3.1 Project Aims

Based on the knowledge gaps identified in the preceding chapter, this project has the following aims:

- Train a centralised ResNet classifier on 20,000 images across 30 classes to establish an upper-bound benchmark
- Run FedAvg and FedProx on the same data, partitioned into IID and Dirichlet non-IID splits across simulated clients
- Report how each federated method degrades from the centralised baseline as heterogeneity increases, and what it costs in communication rounds

3.2 Experimental Pipeline

This project follows a three-stage experimental pipeline: centralised baseline training, federated learning implementation, and comparative evaluation. All experiments will be implemented in Python using PyTorch.

3.2.1 Dataset

Two publicly available skin disease image datasets from Kaggle will be used. The first, the Skin Diseases and Cancer Comprehensive Dataset, contains approximately 13,000 images across 24 diagnostic classes [23]. The second, a larger balanced skin disease dataset, contains approximately 262,000 images across 34 classes [24]. Table 3.1 summarises the two datasets. The final dataset composition will be determined during implementation; options include using either dataset independently, merging them with a unified class mapping, or using a subsampled combination to manage compute requirements. Images will be resized to a uniform resolution and normalised to standardise input across all

experiments. Standard data augmentation techniques such as random horizontal flipping, rotation, and colour jittering will be applied during training to improve generalisation. The dataset will be split into training, validation, and test sets, with the test set held centrally and kept consistent across all experiments to ensure fair comparison.

Table 3.1: Summary statistics of the datasets used in this project.

Property	**Dataset 1**	**Dataset 2**
Name	Skin Diseases + Cancer	Massive Skin Disease
Source	Kaggle	Kaggle
Total images	∼13,000	∼262,000
Number of classes	24	34
Image type	Clinical photographs	Clinical photographs
Class balance	Imbalanced	Balanced

3.2.2 Centralised Baseline

A ResNet-18 architecture pre-trained on ImageNet will be used as the primary classification model across all experiments, selected for its established performance in image classification tasks [25]. If compute resources permit, ResNet-50 will also be evaluated to assess the impact of model capacity on classification performance. The baseline will be trained on the full training set in a standard centralised setting, with all data aggregated on a single machine. This represents an upper bound on classification performance. Training will use stochastic gradient descent with momentum, with learning rate, batch size, and number of epochs tuned on the validation set. To account for variability, experiments will be repeated across multiple random seeds and results reported as mean and standard deviation.

3.2.3 Federated Learning

The training set will be partitioned across K simulated client nodes to replicate a distributed healthcare environment. Two partitioning strategies will be used. In the IID setting, data will be distributed uniformly at random across clients, ensuring each client holds a roughly representative subset. In the non-IID setting, a Dirichlet distribution with concentration parameter α will be used to create heterogeneous label distributions across clients, where lower values of α produce greater skew [13]. Multiple values of α (for example, $\alpha \in \{0.1, 0.5, 1.0\}$) will be tested to evaluate model behaviour under varying degrees of heterogeneity. The inherent class imbalance in the dataset will be considered when designing the partitioning, as it interacts with the label skew introduced by non-IID splits.

FedAvg and FedProx will each be implemented following the procedures described in Chapter 2. The same ResNet-18 architecture and preprocessing pipeline used in the

centralised baseline will be applied to ensure that differences in performance are attributable to the training method rather than the model.

In each communication round, a fraction $C = 0.5$ of $K = 10$ clients will be selected to participate. Each selected client will train on its local data for $E = 5$ local epochs with batch size $B = 32$, then send its updated model parameters to the central server for aggregation.

For FedProx, the proximal term weight μ will be tuned over $\mu \in \{0.001, 0.01, 0.1, 1.0\}$ using the validation set. This allows assessment of its impact under different levels of regularisation.

Training will run for up to $T = 100$ communication rounds. Early stopping will be applied based on validation performance.

3.2.4 Evaluation

All approaches will be evaluated on the same held-out test set using the following metrics.

- **Classification accuracy**: overall accuracy on the test set.
- **Per-class performance**: precision, recall, and F1-score per class to assess behaviour on minority classes, which is particularly important given dataset imbalance.
- **Communication efficiency**: the number of communication rounds required to reach a target accuracy and the total volume of parameter data transmitted between clients and server.
- **Non-IID sensitivity**: accuracy as a function of the Dirichlet concentration parameter α, measuring how each method degrades under increasing data heterogeneity.

Results from FedAvg and FedProx will be compared against each other and against the centralised baseline across all metrics. All experiments will be repeated over multiple random seeds to ensure reported differences are statistically meaningful.

3.3 Ethical Considerations

This project uses only publicly available skin disease image datasets obtained from Kaggle. Both datasets have been released by their authors for research and educational purposes, and contain no directly identifying patient information such as names or dates of birth. No new patient data will be collected, and no human or animal subjects will be involved in the research. For these reasons, formal ethics approval is not required for this project.

Several ethical considerations remain relevant. Publicly released medical image datasets may still carry risks of re-identification. This project will treat all data as sensitive and will not attempt to link images back to individuals or external sources.

Skin disease classifiers are known to exhibit performance disparities across skin tones, as publicly available datasets are often skewed toward lighter skin types. Any results produced by this project will be interpreted with this limitation in mind and per-class performance metrics will be reported to surface any systematic disparities that arise.

While this project is a simulation and does not involve real clinical deployment, the broader motivation carries downstream implications for patient privacy, model accountability, and the trustworthiness of clinical AI systems. This project is a methodological evaluation and no diagnostic claims follow from it.

All code, experimental configurations, and results will be managed with version control to support reproducibility, and the datasets used will be properly cited so that others can verify and build upon the work.

Chapter 4

Study 1: Centralised Baseline and IID Federated Learning

4.1 Aims

- Establish centralised ResNet-18 baseline performance on the 30-class dataset.
- Evaluate FedAvg and FedProx under IID partitioning to verify that federated training approaches the centralised upper bound when data is homogeneous.

4.2 Methods

- Centralised training as described in Chapter 3, with hyperparameters tuned on the validation set.
- FedAvg and FedProx with $K = 10$ clients, $C = 0.5$, $E = 5$, $B = 32$, uniform random data partitioning.

4.3 Results

- To be completed: accuracy and per-class metrics for the centralised baseline.
- To be completed: federated accuracy versus communication rounds under IID conditions.

4.4 Discussion

- To be completed: gap between federated and centralised performance under IID conditions.

Chapter 5

Study 2: Federated Learning under Non-IID Conditions

5.1 Aims

- Evaluate how FedAvg and FedProx degrade as data heterogeneity increases.
- Assess whether the FedProx proximal term improves stability and accuracy under label skew.

5.2 Methods

- Dirichlet partitioning with $\alpha \in \{0.1, 0.5, 1.0\}$.
- FedProx proximal weight tuned over $\mu \in \{0.001, 0.01, 0.1, 1.0\}$.

5.3 Results

- To be completed: accuracy as a function of α for both methods.
- To be completed: communication rounds to target accuracy, per-class performance on minority classes.

5.4 Discussion

- To be completed: comparison of FedAvg and FedProx degradation, practical implications for federated medical imaging.

Chapter 6

General Discussion and Conclusions

6.1 Summary

- To be completed: summary of baseline, IID, and non-IID findings.

6.2 Discussion and Future Work

- To be completed: how the two studies together answer the project aims.
- Planned future directions: stronger architectures such as Vision Transformers, non-IID improvements, communication optimisations.

6.3 Conclusions

- To be completed: overall assessment of whether federated learning is a viable substitute for centralised training in this setting, and under what heterogeneity conditions.

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Appendix A

Program listings

Code and experimental configurations will be made available via a GitHub repository, with directions provided in the final thesis.